

Karan Taneja

✉ karantaneja@gatech.edu
📄 krntneja.github.io

My Introduction: PhD Candidate in CS at Georgia Tech (2021-26), working on conversational AI agents for education. Expert in applied AI including conversational AI, multimodal AI, and NLP. Researcher and engineer passionate about building tools, deploying in production, and performing experiments that measure true value and impact. Experienced teacher, mentor, and leader.

Education

- **Georgia Institute of Technology** Jan'21 – Jun'26 (Expected)
PhD+MS in Computer Science, Grade: 4.00 /4
Atlanta, US
Advised by [Prof. Ashok Goel](#), Thesis: 'Conversational AI for Education — Design, Evaluation and Continual Improvement'
- **Indian Institute of Technology (IIT) Bombay** Jul'15 – Jun'20
BS+MS (Dual Degree) in Electrical Engineering, Grade: 9.63 /10
Mumbai, India
Minor in Computer Science and Engineering

Research and Work Experience

- **Georgia Institute of Technology | Graduate Research Assistant** Atlanta, US
Multimodal Conversational AI, LLMs, and Machine Learning
Mar'21 – Present
LLM-based Virtual Teaching Assistants: [Jill Watson](#) and [MuDoC](#) (See [demo on YouTube](#))
 - ◇ Developed an **LLM-based document-grounded AI agent** for classroom activities using a Flask+Node.js+ReactJS stack.
 - ◇ Created a conversational AI agent based on Reason-and-Act framework for **interleaved text-and-image responses**. ([AIED'25](#))
 - ◇ Showed 45% **higher response acceptance** with improved robustness & low toxicity compared to OpenAI Assistants. ([AIED'24](#))
 - ◇ Studied **learning outcomes** to show that multimodal AI outperforms text-only AI on both outcomes and experience. ([AIED'26](#))
 - ◇ Used by **6K+ students across multiple institutions**, showed improvements in student perception. ([ITS'24](#), [L@S'24](#))Machine Teaching and Active Label Correction ([ALC3](#))
 - ◇ Built ALC data and training pipelines to **improve LLM-based AI agents with human-in-the-loop** using deployment data.
 - ◇ Matched oracle accuracy with a fraction of annotations, leading to a **16% gain over existing ALC baselines**. ([EMNLP'24](#))
 - ◇ Designed a framework for knowledge-aided machine teaching to support **collaborative annotation**. ([CHAI-IJCAI'22](#))Other Collaborative Projects
 - ◇ Led the [AWS deployment of MuDoC](#) as Jill Watson in classrooms at Georgia Tech, making it available on Canvas LMS.
 - ◇ Developed [MetaCues](#), a **conversational AI that provides metacognitive guidance** based on interaction logs (UT Austin).
 - ◇ Developed [HALT](#), a **lightweight hallucination detector** that leverages LLM log probabilities as a time series sequence ([arXiv](#)).
 - ◇ Building a benchmark for **evaluating document-grounded interleaved text-and-image** retrieval and response generation.
- **IBM Thomas J. Watson Research Center | Research Intern (Twice)** New York, US
Computational Creativity with Large Language Models
Long-form Story Generation using Cumulative Summarization and RL May'23 – Aug'23
 - ◇ Orchestrated a Flan-T5 **summarizer-generator loop with cumulative summarization** for long-form story generation.
 - ◇ Trained models with a **ZeRO-optimized distributed RL pipeline** for training models on Project Gutenberg stories. ([Poster](#))Constrained Recipe Generation using LLMs and Monte Carlo Tree Search May'22 – Aug'22
 - ◇ Trained GPT-2 LLM for food recipe generation and developed **RecipeMC inference method for imposing constraints**.
 - ◇ Employed **Monte Carlo Tree Search to sample tokens** based on joint LLM outputs and constraint-defining reward functions.
 - ◇ RecipeMC **ingredients & instructions were preferred over ground truth** 55% & 62% of the time respectively. ([ICCC'23](#)).
- **Microsoft India Development Center + IIT Bombay | Research Intern and Collaborator** Hyderabad, India
Code-mixed (CM) Hindi-English Automatic Speech Recognition (ASR) Oct'18 – Feb'20
 - ◇ Developed **linguistically-grounded algorithms** to create synthetic CM English-Hindi corpora for low-resource ASR training.
 - ◇ **Lowered error rate for English-Hindi ASR** by 5% using these synthetic corpora along with a code-mixed corpus. ([IS'19](#))
 - ◇ Improved ASR performance using **text-to-speech generations & loss functions design** to promote code-switching. ([IS'20](#))
- **IIT Bombay | Graduate Research Assistant** Mumbai, India
fMRI Acquisition and Reconstruction using Convolutional Neural Networks (CNNs) Jan'19 – Jun'20
 - ◇ Engineered a **multi-stage CNN pipeline for reconstruction** using noisy under-sampled frequency-domain R-fMRI acquisition.
 - ◇ Used **Bayesian uncertainty estimation in CNN outputs** to get insights into reconstruction quality of brain activity networks.
 - ◇ Enabled **8x higher spatial resolution** without compromising temporal resolution or increasing scan time. ([ICPR'20](#)).
- **Sony Semiconductor Solutions | Research Intern** Atsugi, Japan
Spherical CNN CUDA Kernels for human detection task in 360-degree images May'18 – Jul'18
- **Philips Innovation Campus | Software Intern** Bangalore, India
Patient Motion Detection during MRI Scans using hybrid CNN-RNN models May'17 – Jul'17
- **Teaching Experience**
Knowledge-based AI (TA at Georgia Tech), AI for High-school Students (Gujarat), Machine Learning (TA at IIT Bombay), Data Analysis and Interpretation (TA at IIT Bombay, Twice), Web Development for High-school Students (Mumbai)

Selected Publications [[Google Scholar](#), 250+ citations, h-index: 7]

- **Impact of Multimodal and Conversational AI on Learning Outcomes and Experience** [[pdf](#)]
K. Taneja, A. Singh, A. Goel. International Conference on Artificial Intelligence in Education (AIED) 2026.
- **Towards a Multimodal Document-grounded Conversational AI System for Education** [[pdf](#)]
K. Taneja, A. Singh, A. Goel. International Conference on Artificial Intelligence in Education (AIED) 2025.
- **MuDoC: An Interactive Multimodal Document-grounded Conversational AI System** [[pdf](#), [demo](#)]
K. Taneja, A. Goel. AAAI Symposium on Machine Learning and Knowledge Engineering (AAAI-MAKE) 2025.
- **Can Active Label Correction Improve LLM-based Modular AI Systems?** [[pdf](#), [video](#)]
K. Taneja, A. Goel. Conference on Empirical Methods in Natural Language Processing (EMNLP) 2024.
- **Jill Watson: A Virtual Teaching Assistant powered by ChatGPT** [[pdf](#)]
K. Taneja, P. Maiti, et al. International Conference on Artificial Intelligence in Education (AIED) 2024.
- **Monte Carlo Tree Search for Recipe Generation using GPT-2** [[pdf](#), [slides](#)]
K. Taneja, R. Segal, R. Goodwin. International Conference on Computational Creativity (ICCC) 2023.
- **Exploiting Monolingual Speech Corpora for Code-mixed ASR** [[pdf](#), [poster](#)]
K. Taneja, S. Guha, P. Jyothi, B. Abraham. Interspeech 2019 (ISCA Annual Conference).

Technical Skills

Applied AI Research	Conversational AI, Large Language Models, Multimodal AI, AI in Education, Natural Language Processing, Deep Learning, Experiment Design, User Studies, Prototyping, Cloud Deployment
Programming	Python (primary), C/C++, Java, Full-stack Web Development
Tools	ML+LLMs: HuggingFace, Transformers, LangGraph, SpaCy, NLTK, PyTorch, TensorFlow, Scikit-Learn, Keras Data Analysis: NumPy, Pandas, SciPy, Matplotlib Backend: Flask, Django, Node.js, MongoDB, Weaviate, AWS, GCP, OpenAI, Docker, Nginx, Locust, Linux, Bash Scripting Prototyping: React JS, Material UI, Streamlit User Study: Prolific, Qualtrics, Surveys, Interviews

Leadership, Service, Awards

- **Asha for Education – US-based 501(c)(3) Philanthropic Organization**
Board Member, Asha-wide Treasury Team Jun'24 – Jun'25
 - ◊ Worked as a part of 6-member Treasury Team to manage finances, USD 4MM+ in annual revenue and 8MM+ in assets.
 - ◊ Spanned across 40+ chapters around the world and 200+ NGOs, touching lives of 200K+ students every year.*Co-President, Atlanta Chapter* May'23 – Apr'24
 - ◊ Led a team of 30+ volunteers as one of two co-presidents to organize fundraising events, festival celebrations, and outreach.
 - ◊ Worked on community building through regular socials, virtual site-visits to NGOs in India, and a strong social media presence.*Treasurer, Atlanta Chapter* Aug'22 – Apr'23
 - ◊ Managed Atlanta chapter funds that supported six NGOs in India with an annual philanthropic giving budget of USD 120K.
 - ◊ Led fundraising efforts through 5K/10K/HM Run Training Program, social events, concessions, and grant applications.
- **Academic Reviewing and Mentoring**
 - ◊ Reviewer for ACL Rolling Reviews 2023, AAAI Special Award Judge at ISEF 2022, ICPR 2020
 - ◊ Mentored a masters student on hallucination prediction in LLMs (Georgia Tech)
 - ◊ Mentored two masters students on deploying a multimodal conversational agent on cloud (Georgia Tech)
 - ◊ Mentored an undergraduate student in building a conversational AI with metacognitive cues (UT Austin)
 - ◊ Mentored three freshmen in the 'Summer of Science' projects on introductory ML and AI (IIT Bombay)
- **Invited Talks and Panels**
 - ◊ Worcester Polytechnic Institute: [CS 565 User Modelling](#) (October'25), Host: Prof. Neil Heffernan
 - ◊ [Digital Education Dialogues Podcast Series](#), The Digital School (September'25) | Host: Prof. Christopher Dede
 - ◊ Panel on "AI, Teaching, Learning, and Job Readiness", Georgia Tech (April'25) | Host: Prof. Olufisayo Omojokun
 - ◊ [Intelligence and Computing Education Lab](#) at Utah State University (March'25) | Host: Prof. Yang Shi
 - ◊ Frontiers Research Talks ([video](#)), [Manifold Research](#) (December'24) | Host: Harshvardhan Sikka
 - ◊ Worcester Polytechnic Institute, [CS 568 AI for Adaptive EdTech](#) (October'24) | Host: Prof. Neil Heffernan
- **Awards and Honors**
 - ◊ Awarded with Institute Academic Prize for ranking 1st in EE department, IIT Bombay for the year 2018-19.
 - ◊ Awarded with Technical Color Prize (H2) for contribution to inter-hostel tech competitions at IIT Bombay.
 - ◊ Awarded with International Speech Communication Association Travel Grant to attend InterSpeech 2019 in Graz, Austria.
 - ◊ Kishore Vaigyanik Protsahan Yojana (KVPY) Fellow 2014, invited to National Science Camp at IISER Kolkata.
 - ◊ National Talent Search Scheme (NTSS) Fellow 2013, awarded yearly stipend for undergraduate education.